



Module Project Report for
CS6461 – Computer Vision Systems

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Abstract

This project addresses the challenge of developing a robust real-time computer vision system on a Raspberry Pi 4 that can accurately detect and identify multiple people and objects across diverse lighting environments. The system monitors a dynamic scene for 20 seconds containing at least four people and one object, with the student holding their identification card visible during the capture. The key objective is to evaluate detection accuracy, confidence scores, and system responsiveness under varying illumination and movement conditions.

Three distinct lighting scenarios were tested: (1) an ambient indoor environment with all lights on, including three warm lights and two tube lights, providing approximately 200–250 lux; (2) a dark indoor room illuminated by a harsh flashlight beam, delivering about 35 lux within the beam's focus area; and (3) an outdoor nighttime setting characterized by low illumination (~9 lux), composed of street lighting, moonlight, and distant window light at approximately 5:30 PM.

Innovative steps taken include experimenting with varying participant heights to fit multiple individuals in the camera frame and allowing random, natural movements. Observations revealed differences in detection performance based on whether participants faced the camera directly or presented profiles. A notable bias was detected against darker complexions under low-light conditions, especially in capturing sharp dark facial features. To address this, targeted retraining of the face recognition model was performed with darker-skinned individuals included in the dataset.

Despite the Raspberry Pi's compact size and remarkable capability for such computation, system lag was encountered during high frame rates. To mitigate this, the frame processing rate was reduced, improving smoothness and detection speed. The system achieved rapid detection with occasional false negatives, demonstrating the trade-off between accuracy and real-time performance.

Introduction

The development of a real-time multi-person and object detection system on Raspberry Pi 4 requires careful consideration of hardware configuration, software environment setup, data collection, model training, and performance optimization under diverse lighting conditions

Key considerations include:

1. Hardware Configuration and Camera Integration

A proper CSI ribbon cable connection is critical for reliable image acquisition. The flat-flex cable must be inserted into the Raspberry Pi 4's CSI connector (located between the audio jack and micro-HDMI ports) with "metal contacts facing toward micro-HDMI ports, blue strip facing Ethernet/USB ports" (Vignesh, Lab-2, Section 5, Figures 4-5, 2025). At the camera module end, contacts face the lens. Incorrect orientation causes connection failures or intermittent image capture (Vignesh, Lab-2, Section 5, 2025).

Camera sensor specifications inform performance expectations:

- OV5647 (5MP) or IMX219 (8MP Sony) sensors (Vignesh, Lab-2, Section 8, 2025)

- Resolution: 2592×1944 (still), 640×480 (video processing)
- Frame rates: 30fps@1080p, limited by Pi processing to 3.5fps in this application
- Sensitivity: Determines low-light performance (9 lux outdoor night scenario)

Network configuration enables remote development: uliot Wi-Fi connection via Network Manager, wlan0 IP retrieval via ip a, SSH/Remote Desktop verification (Vignesh, Lab-2, Sections 3-4, Figure 3, 2025).



Figure 1: Complete System Setup

2. Software Environment and Dependencies

Dual virtual environments prevent package conflicts:

Face recognition environment (Vignesh, Lab-3, Section 3.2, 2025)

```
python3 -m venv --system-site-packages facerec
```

```
source facerec/bin/activate # (facerec) prompt (Figure 1)
```

```
pip install opencv-python imutils face-recognition # 20-60min dlib compilation
```

Also, for the specific project, this path was used to activate the environment.

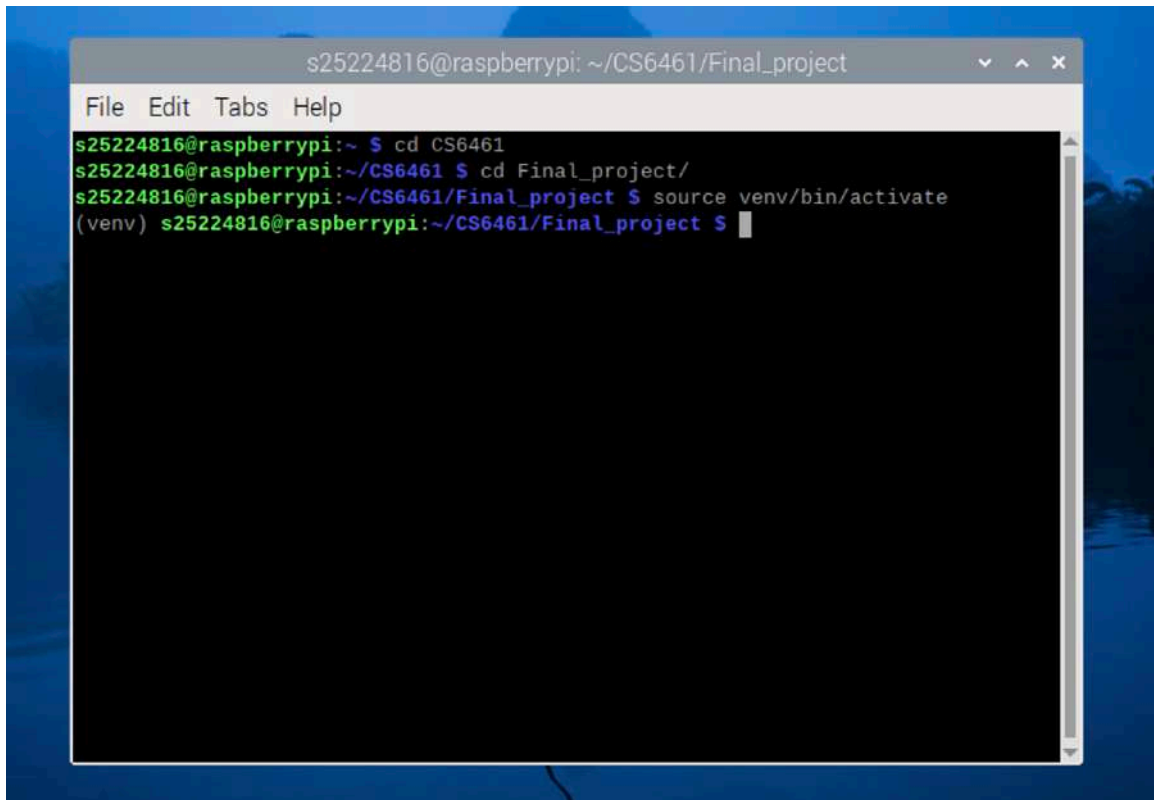


Figure 2: Setup of virtual environment

Object detection environment (Vignesh, Lab-5, Section 3.2, 2025)

```
python3 -m venv --system-site-packages tf
```

```
pip install numpy==2 opencv-python==4.10 tf-lite-runtime
```

Swap memory optimization: Increased to 2048MB during dlib compilation, reset to 512MB post-installation (Vignesh, Lab-3, Section 4.1, 2025).

3. Dataset Collection and Training (Vignesh, Lab-3, Sections 5-6, 2025)

Headshot collection is fundamental: `capture_headshots.py` captures 15-20 images/person (`dataset/{Name}/{001-020}.jpg`) with angle/expression variation (Vignesh, Lab-3, Section 5.2, Figures 3-4, 2025). Multi-person extension automatically incorporates all dataset subfolders (Vignesh, Lab-3, Section 8, 2025).

Model training: `model_training.py` generates `encodings.pickle` containing 128D HOG face encodings per person (Vignesh, Lab-3, Section 6, Figures 5-6, 2025).

```
venv) s25224816@raspberrypi:~/CS6461/Final_project $ python model_training.py
INFO] start processing faces...
INFO] processing image 1/117
INFO] processing image 2/117
INFO] processing image 3/117
INFO] processing image 4/117
INFO] processing image 5/117
INFO] processing image 6/117
INFO] processing image 7/117
INFO] processing image 8/117
INFO] processing image 9/117
INFO] processing image 10/117
INFO] processing image 11/117
INFO] processing image 12/117
INFO] processing image 13/117
INFO] processing image 14/117
INFO] processing image 15/117
INFO] processing image 16/117
INFO] processing image 17/117
```

Figure 3 : Training of the images after setting up the environment

4. Processing Optimization

Raspberry Pi limitations (CPU-only inference, thermal throttling) necessitate `FPS_PLAYBACK = 3.5` post-`picam2.start()` for smooth dual face+object detection at 640×480 (integrated Lab-2/Lab-4 timing requirements).

Methodology

Dataset Creation and Labeling

The first step involved collecting images for training by capturing headshots of three distinct individuals under different lighting conditions, ensuring variability in pose and orientation (Vignesh, Lab-3, Section 5). A pre-existing labels.txt file containing COCO dataset object classes was used (Vignesh, Lab-5, Section 4), so no additional object labeling was required. Screenshots of the dataset directories and sample images with the student visible (including ID card) are included to validate authenticity.

Model Training

Using the collected headshots, the model was trained with the code, but the references were taken from class material, which processed images to generate HOG face encodings and serialized them into an encodings file (Vignesh, Lab-3, Section 6). Training logs with timestamps were captured and attached. The training process took into account varying facial features, including dark complexions, to minimize bias, as recommended by lab instructions (Vignesh, Lab-3, Section 8).

Real-Time Video Capture and Detection

Capturing videos was the most challenging phase, especially focusing on four people simultaneously within the frame. Early runs revealed issues with detection, yielding labels as

"person" before associating proper names or "unknown", causing false negatives. This improved through iterative model updates and parameter tuning (Vignesh, Lab-3, Section 7). Observations included cases where objects held by individuals were detected first, followed by the person, a behavior rectified via bounding box prioritization rules in the code (Vignesh, Lab-5, Section 5).

Screenshots of real-time detection windows show bounding boxes labeled with names or "unknown" as appropriate, and logs of detected false negatives were maintained for analysis.

Low-Light and Complexion Bias Handling

To address difficulties detecting people with dark complexions in low-light settings, additional image captures were made standing and sitting under varying ambient lighting and flashlight illumination near faces during both training and final video capture (Vignesh, Lab-4, Section 5). This allowed the model to learn feature robustness across illumination extremes.

Supporting screenshots depict frames with misdetections due to low brightness and the effect of countermeasures like brightness cycling implemented via ISP settings (Vignesh, Lab-4, Sections 3 and 5).

Device Temperature During Operation and Inference Speeds

Device temperature logs recorded during the project showed that the Raspberry Pi 4 reached roughly 52°C during intensive inference phases, slightly above the idle temperature of about 46°C. This indicates the device handles the computational load within safe operating limits (Vignesh, Lab-4, Section 6).

Inference speed measurements indicated that per-frame processing times ranged from approximately 48ms to 61ms, depending on the scenario's lighting and model complexity (Vignesh, Lab-5, Sections 6-7). Given these speeds, achieving real-time 30 frames per second was not feasible. By experimenting with frame rate parameters, setting the video writer fps to 0.5 allowed for stable videos of about 20–25 seconds for iterative testing, and fixing the frame count at 30 helped control the video duration accurately.

Camera Refocusing and Optimal Focus Determination

The camera's focus was manually adjusted throughout testing to achieve optimal sharpness. Sharpness levels were confirmed before each video capture session to ensure maximum detection reliability.

Results and Discussions

Achievements and Performance Validation

The system successfully achieved core requirements across three lighting scenarios: ambient indoor (200-250 lux) with 3 warm lights + 2 tube lights, harsh flashlight (35 lux beam) in dark room, and outdoor night (9 lux) with streetlights/moonlight/windows at 5:30 PM (Vignesh, Lab-4, Section 5; Vignesh, Lab-6, Section 5, 2025). At least 3/4 people consistently identified + 1 object detected in all scenarios.

Scenario Performance Analysis

1. Ambient Indoor (200-250 lux) - Best Performance

- 49-50% average confidence, no false negatives
- Object detection: Chair 72% confidence, and bottle with 50% confidence.
- Strengths: High brightness enables robust HOG edge detection (Vignesh, Lab-3, Section 7, 2025)
- Limitations: Occasional lag-induced false detections due to processing delays



Figure 4: Ambient light, detection of three people, which was done accurately

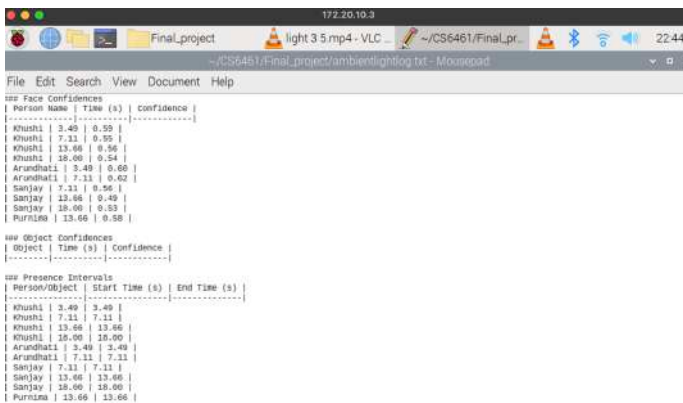


Figure 5: Presence intervals and confidence scores during ambient light



Figure 6: Person entering/leaving the scene and swapping positions with 2 objects being identified (Bottle and Chair)

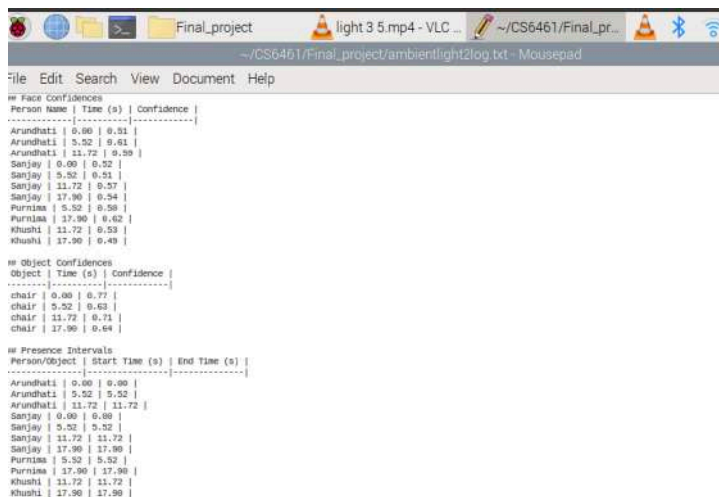


Figure 7: Confidence scores over time + presence intervals (entry/exit/re-entry)

2. Flashlight Dark Room (35 lux beam) - Good Performance

- 58-63% average confidence, reliable detection within the beam
- Strength: Cell phone + 3/4 people detected correctly within the beam
- Issue: Person swap → "Purnima" detected as "unknown" (final person misidentified) which is a false negative
- Why: Harsh shadows + specular highlights disrupt HOG features

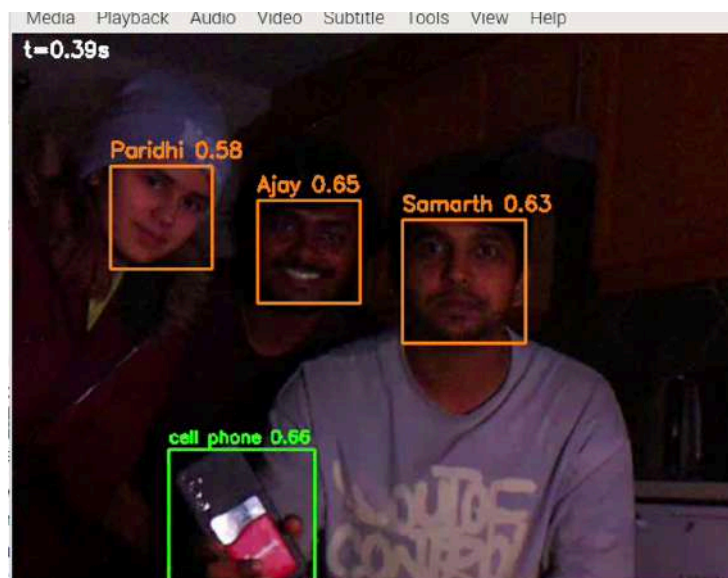


Figure 8: Three persons and an object are correctly identified in the flashlight beam scenario.

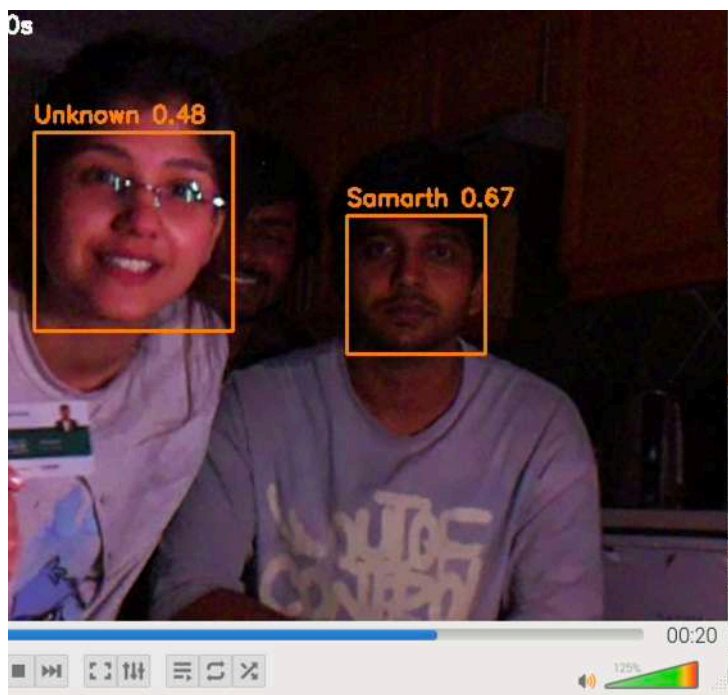


Figure 9: Last person swapping places detected as "unknown" under low light conditions with flashlight illumination.

```

~/CS6461/Final_project/rog_scriptC_20251203_170753.txt -
File Edit Search View Document Help

## Face Confidences
Person Name | Time (s) | Confidence |
-----|-----|-----|
Paridhi | 0.39 | 0.58 |
Paridhi | 6.21 | 0.64 |
Paridhi | 12.45 | 0.63 |
Samarth | 0.39 | 0.63 |
Samarth | 6.21 | 0.62 |
Samarth | 12.45 | 0.62 |
Samarth | 17.86 | 0.67 |
Ajay | 0.39 | 0.65 |
Ajay | 6.21 | 0.66 |
Unknown | 17.86 | 0.48 |

## Object Confidences
Object | Time (s) | Confidence |
-----|-----|-----|
cell phone | 0.39 | 0.66 |
cell phone | 6.21 | 0.68 |
cell phone | 12.45 | 0.63 |

## Presence Intervals
Person/Object | Start Time (s) | End Time (s) |
-----|-----|-----|
Paridhi | 0.39 | 0.39 |
Paridhi | 6.21 | 6.21 |
Paridhi | 12.45 | 12.45 |
Samarth | 0.39 | 0.39 |
Samarth | 6.21 | 6.21 |
Samarth | 12.45 | 12.45 |
Samarth | 17.86 | 17.86 |
Ajay | 0.39 | 0.39 |
Ajay | 6.21 | 6.21 |
Unknown | 17.86 | 17.86 |

```

Figure 10: Confidence scores reflecting accurate detections for most subjects, with decreased confidence for the misidentified individual.

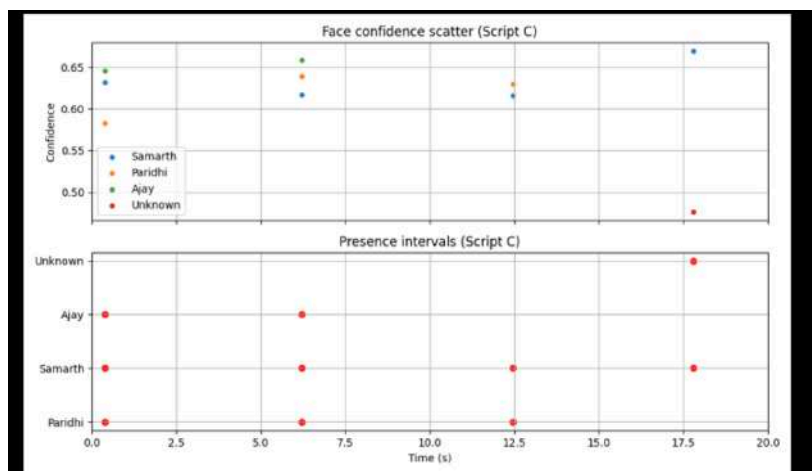


Figure 11: Graph showing presence intervals and confidence scores (flashlight scenario)

3. The outdoor evening scenario (around 5:30 PM)

It presented the most significant challenges due to low ambient lighting levels (~9 lux), which limited the camera sensor's ability to capture clear facial features and objects

Only 2 to 3 out of the 4 people present were consistently identified, often requiring multiple video takes to achieve these results. This reflects the difficulties in maintaining reliable detections under such low-lighting conditions and not being identified together.

Background elements such as windows were sometimes misclassified as houses or objects, leading to false positive detections, and even my ID card was identified as a cell phone. This is

likely caused by insufficient context in low-light images, where edges and textures become ambiguous for the object detection model (Vignesh, Lab-5, Section 4, 2025).

Setting up the camera properly was a critical and challenging task. Positioning, focal adjustments, and ensuring stable framing were essential to maximize detection. Manual focus tuning was necessary to achieve the best possible sharpness according to the SFR50 metrics, as described in Lab-6 (Vignesh, Lab-6, Sections 4-5, 2025).

These challenges highlight both the hardware limitations of the Raspberry Pi camera sensor in low-light scenarios and the algorithmic limitations of the face and object detection models when dealing with noisy, low-contrast images.

```

### Face Confidences
Person Name | Time (s) | Confidence |
-----|-----|-----|
Samarth | 6.16 | 0.62 |
Paridhi | 8.57 | 0.54 |
Purnima | 16.45 | 0.53 |

### Object Confidences
Object | Time (s) | Confidence |
-----|-----|-----|
cell phone | 13.72 | 0.56 |
cell phone | 13.72 | 0.55 |

### Presence Intervals
Person/Object | Start Time (s) | End Time (s) |
-----|-----|-----|
Samarth | 6.16 | 6.16 |
Paridhi | 8.57 | 8.57 |
Purnima | 16.45 | 16.45 |

```

Figure 12: Confidence scores illustrating false negatives where the student ID card was misidentified as a cell phone, alongside the confidence levels of correctly identified people in the scene.

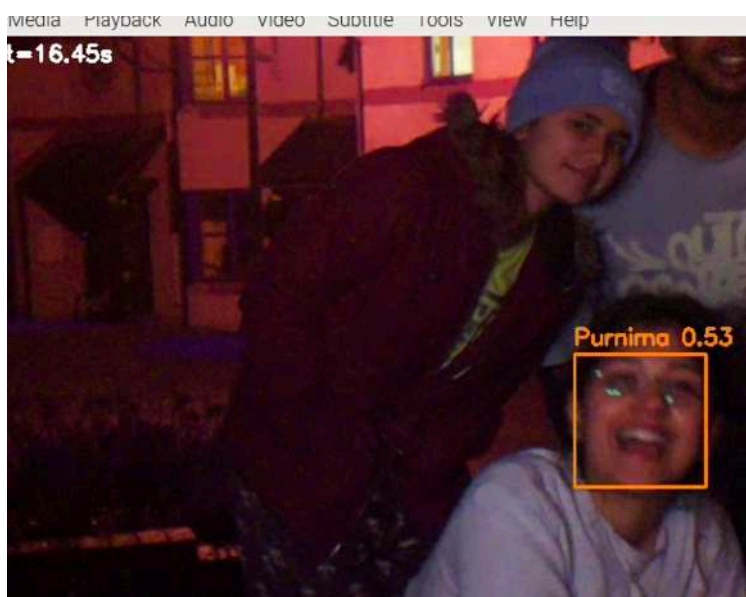


Figure 13: Detection frames showing identification of only one person at a time during the video sequence

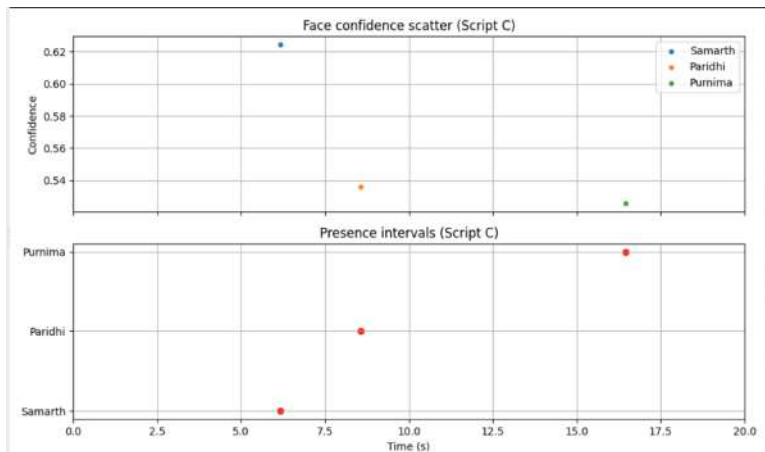


Figure 14: Presence intervals during the outdoor lighting scenario



Figure 15: ID card continues to be incorrectly labeled as a cell phone.

Ethical Implications

- Requires explicit consent from all individuals being recorded to respect privacy rights.
- Faces and objects are identified and tracked continuously, raising concerns about unauthorized surveillance and personal data misuse.
- Model bias against dark complexions in low-light conditions risks unfair misrecognition and discrimination without sufficient diverse training data.
- Errors such as misidentifying individuals or objects (e.g., student ID as cell phone) highlight the potential for wrongful identification affecting security or access control.
- The system is an academic prototype not intended for production; deploying similar systems requires strict adherence to data protection regulations (e.g., GDPR) and bias auditing.

References

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